

Tech Gadget Store

1. Functional & Non-Functional Requirements

Functional Requirements

- **Product Display:** The page must display a grid of products with images, titles, prices, and categories.
- **Search Functionality:** Users should be able to type in a search bar to filter products by name in real-time.
- **Category Filtering:** Users should be able to click category buttons (e.g., "Laptops", "Audio") to filter the list.
- **Responsive Layout:** The catalogue should adjust from a multi-column grid on desktop to a single column on mobile.

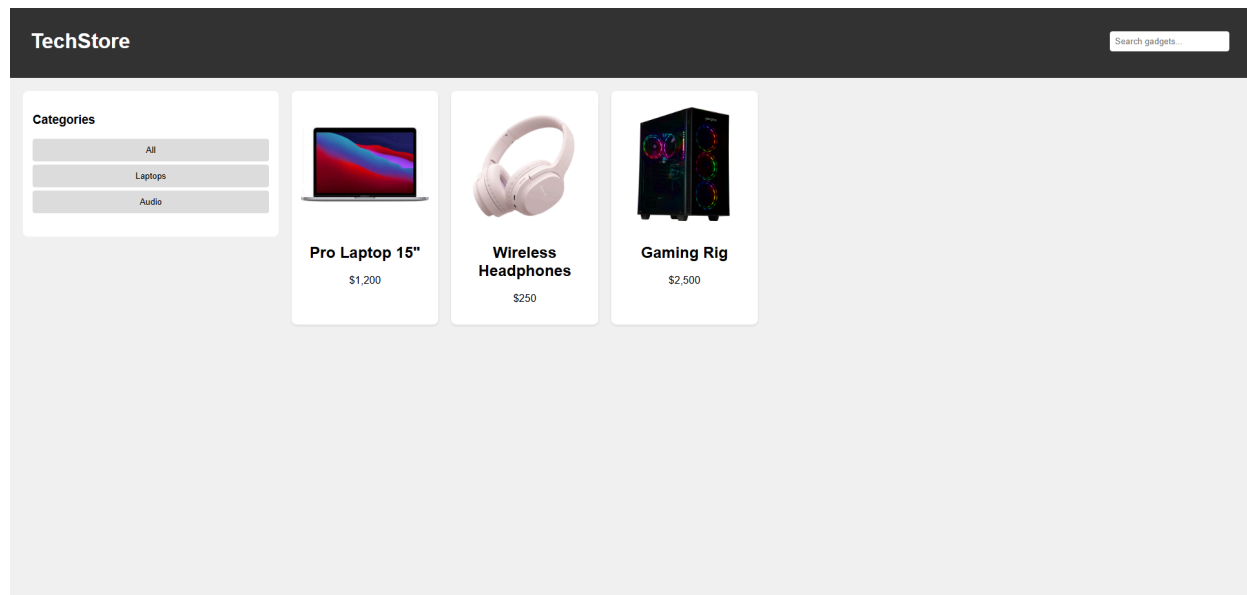
Non-Functional Requirements

- **Performance:** The page should load quickly by using optimized image placeholders.
 - **Accessibility:** Use semantic HTML (like `<main>`, `<article>`, and `aria-labels`) to ensure screen readers can navigate the catalogue.
 - **Maintainability:** CSS must be separated into an external file with a clear naming convention (BEM or similar).
-

2. Page Design (Schema)

The design follows a standard E-commerce layout to ensure it is "unambiguous."

- **Header:** Contains the site branding and a centered search input.
- **Sidebar (Aside):** A vertical list of category filters.
- **Main Content:** A responsive CSS Grid containing product "Cards." Each card includes an image, a heading, and a price tag.



3. Implementation Strategy

Semantic HTML

Instead of using `<div>` for everything, we will use:

- `<header>` for the top navigation.
- `<aside>` for the filter sidebar.
- `<main>` for the primary content area.
- `<article class="product-card">` for individual items.

External CSS

We will use an external `style.css` file. To ensure points for "mapping to HTML using classes," we will avoid styling raw tags and instead use classes:

CSS

```
/* Example */
.product-grid {
  display: grid;
  grid-template-columns: repeat(auto-fill, minmax(250px, 1fr));
  gap: 20px;
}

.product-card {
```

```
border: 1px solid #ddd;  
padding: 15px;  
border-radius: 8px;  
}
```

JS Events for Functionality

To meet the requirement of "affecting functionality," we won't just use JS for an alert. We will use it to **filter the DOM**:

- **Event:** `keyup` on the search bar.
- **Function:** Iterates through all `.product-card` elements and sets `display: none` if the title doesn't match the search string.